How we added explicit sleep

Single-qubit memory experiment | Aqora / Quantinuum H2-1E | Team explanation

The sleep instruction was written directly into QIR, not Guppy. A local Python script generates the QIR text. LLVM/Clang assembles it into .bc bitcode, which is submitted to H2-1E through Aqora.

1. The one-qubit memory circuit

Prepare |0> → Ry(π/2) → sleep(t) → Ry(-π/2) → measure Z

The first rotation prepares |+>. The second converts an X-basis measurement into a Z-basis measurement, making the probe sensitive to dephasing during the wait. Without errors, the recorded bit is 0; a recorded 1 is an X-memory error. The rotations are implemented using native phased-X QIR calls.

2. The exact sleep instruction

```
declare void @__quantum__qis__sleep__body(%Qubit*, double)

; Example: raw sleep argument 0.5 on virtual qubit 0
call void @__quantum__qis__sleep__body(
    %Qubit* null,
    double 5.00000000000000000e-01
)
```

Under the static QIR convention, `%Qubit* null` identifies **virtual qubit 0**; it does not mean “no qubit.” The backend assigns the physical ion. The `double` is the raw sleep argument. The declaration specifies the function signature; the call applies it between the two rotations.

3. How the sweep was executed

Five otherwise identical programs used explicit sleep arguments **0, 0.1, 0.5, 0.75 and 1**, with 1,000 shots each. The zero-duration control also contains a sleep call. Each program retained both rotations and its requested sleep argument after assembly and bitcode round-trip, and all passed the H2-1SC syntax checker.

The jobs requested noisy stabilizer simulation, idle and transport dephasing enabled, and optimization disabled (`no_opt=true`) to preserve the sequence. **No artificial errors were injected.** We measured the emulator output after supplying the explicit sleep instruction.

4. What this establishes

All five H2-1E jobs completed. The measured error counts were **0, 1, 21, 55 and 92 out of 1,000**, respectively. This shows a clear dependence on the sleep argument in this probe. It does **not** independently establish the argument’s physical units, a calibrated scaling law, or the backend’s internal noise implementation.

Implementation and saved evidence

`SG_final_round/src/single_qubit_pilot.py` - reusable QIR template.
`SG_final_round/src/single_qubit_sweep.py` - build, submit, fetch and analyze.
`SG_final_round/results/single_qubit_sweep_1000.json` - counts and job IDs.